# Public-Data Hedging Benchmark

Approach, current results, and model catalogue

Status date: 13 October 2026

**Executive status.** The model-neutral daily benchmark is operational on a normalized, original-data-free public-candidate panel. The current target-notional cohort has complete out-of-sample results for SANOS–LV and the three robust-covariance mean–variance policies over all 611 eligible 2023–2025 origins and all 20 daily digital targets. SANOS–LV reduces aggregate pure-hedge RMSE by **56.61%**; the completed covariance policies reduce it by **7.62%–20.79%**, while Historical-SANOS mean–variance at $\lambda = 0.1$ reduces it by **38.19%**. The remaining Historical-SANOS and covariance objectives are still running. The 611-origin GROMM preparation is complete and its rolling Q fit is running on the managed GPU. VolGAN daily readiness passes and corrected frozen-weekly readiness is queued behind GROMM; AD–Seq–Vol and its fine-tuned variant remain behind the same authenticated readiness gate. Weekly mechanics and pricing pass, but full weekly production has not yet started.

## 1 Approach

### 1.1 Data boundary and common market

The sole runtime input is the distributable-candidate dataset `spx-2022-2025-hedging-public-candidate-v1`. It contains 891 retained dates from 3 January 2022 to 29 August 2025 and 2,014,740 contemporaneously eligible quote-side option coordinates. It contains no original WRDS price, midpoint, identifier, volume, open interest, vendor Greek, or vendor implied volatility. Instead it releases normalized fitted SANOS and DLV states, separate reconstructable SANOS and DLV model prices, Black implied volatilities, forward/spot ratios, discount factors, normalized half-spreads, and forward-adjusted spot returns.

The common executable market mark is the released SANOS price. A purchase pays SANOS plus one half-spread and a sale receives SANOS minus one half-spread. Spot costs one basis point of traded notional per transaction; cash settlement is free. The release is technically classified as `public_candidate`; redistribution still requires a separate rights decision.

### 1.2 Out-of-sample protocol

- Model initialization, architecture validation, and target sizing use 2022 only. At each date $t$, every rolling estimator uses the current market state and the trailing 252 exact one-session transitions available at $t$, with no future observation.
- Daily evaluation runs from 17 January 2023 through 28 August 2025: 611 eligible origins, 20 targets per origin, and 12,220 claim/origin records.
- The targets are cash digitals at 1D and 3D: ATM call and put plus OTM NMS levels $\pm 0.25, \pm 0.50, \pm 0.75, \pm 1.00$. Quantities are fixed from 2022 so each target has one maturity-matched ATM spot-risk unit.
- The hedge universe is normalized spot plus every eligible released market strike strictly inside $(-2, 2)$ contemporaneous NMS, represented by its OTM option (put below the forward and call at or above it). It uses every expiry up to five business days, or the first later expiry when a five-day expiry is absent.
- Weekly evaluation starts an overlapping five-session episode on every eligible date. Frozen weekly fits once and keeps its inception contract book; restarted weekly refits daily. Both may react to the realized state.

### 1.3 Common accounting and objectives

Every method starts from the same market claim mark and is evaluated on the same realized next-date SANOS mark or cash payoff. The report separates *pricing loss*, *pure hedge loss*, and their sum, *economic loss*. Buyer and seller indifference prices are additional model outputs, not substitutes for this common-price comparison.

Supported risk criteria are mean–variance, mean–volatility, entropy OCE, and power-exponential OCE, following the risk-functional hedging framework of Buehler et al. [ ]. Learned or statistical policies use the hedge of the claim relative to the corresponding zero-claim policy; generated models also fit a direct pure policy so transaction costs are charged on the executable net position rather than inferred from differences of absolute-policy costs.

## 2 Current Results

### 2.1 Authenticated current cohort

Table 1 contains only policies with the same public dataset manifest and fixed 2022 target-notional identity. Every row contains all 611 eligible origins and 12,220 claim/origin records, with zero position-cap hits and zero nonconvergence.

The common unhedged RMSE is 163.16 basis points. "Reduction" is relative to that value. Pure $ES_{95}$ is the equal-weighted mean of target-level expected shortfalls. Economic book MDD is the maximum drawdown of the cumulative P&L of an equal-weighted 20-target daily book. Losses, costs, and drawdowns are in basis points of inception normalized spot; gross exposure is normalized entry notional.

Table 1: Authenticated current-cohort daily results.

| Policy | Pure RMSE | Reduction | Pure $ES_{95}$ | Mean cost | Gross exposure | Economic book MDD |
|---|---|---|---|---|---|---|
| SANOS–LV reference | 70.80 | 56.61% | 134.71 | 10.10 | 2.6640 | 6,779.67 |
| Historical SANOS MV, $\lambda = 0.1$ | 100.85 | 38.19% | 179.07 | 2.45 | 0.0236 | 2,752.93 |
| Covariance MV, $\lambda = 0.1$ | 150.73 | 7.62% | 215.13 | 0.66 | 0.0048 | 1,723.98 |
| Covariance MV, $\lambda = 1$ | 135.85 | 16.74% | 209.59 | 0.74 | 0.0062 | 1,720.29 |
| Covariance MV, $\lambda = 10$ | 129.25 | 20.79% | 257.48 | 1.48 | 0.0518 | 1,754.14 |

SANOS–LV gives the largest RMSE reduction in this completed subset, but it uses a materially larger marked book and pays more transaction cost. The covariance policies trade smaller books and have smaller cumulative book drawdown. Historical-SANOS mean–variance at $\lambda = 0.1$ ranks second by both RMSE and $ES_{95}$ in the completed subset while using much less gross exposure than SANOS–LV. The $\lambda = 10$ covariance policy has its best RMSE, whereas its tail expected shortfall is worse than at $\lambda = 1$. The remaining fitted objectives must complete before this becomes a full model ranking.

### 2.2 Evidence level across models

Table 2: Current evidence. "Readiness" is not a performance estimate.

| Model | Measure | Daily status | Evidence |
|---|---|---|---|
| SANOS–LV | Q price / realized path | **complete** | Current cohort: 611 origins, 12,220 records, zero cap hits and zero nonconvergence. |
| Historical SANOS | P | **pending** | Mean–variance $\lambda = 0.1$ is complete on the current cohort; the remaining daily objectives are running. Exact-cost weekly entropy and power first-origin gates pass, while full weekly production remains pending. Superseded-cohort results are excluded. |
| Robust covariance | P | **pending** | Mean–variance $\lambda \in \{0.1, 1, 10\}$ is complete on the current cohort; mean–volatility production is running. |
| GROMM | Q only | **ready** | All 611 dated inputs are prepared; rolling Q fitting is running on the managed GPU. Compatible P remains rejected by its fixed gates. |
| VolGAN | P | **ready** | Daily public readiness passes; corrected frozen-weekly public readiness is queued behind GROMM. Full production remains pending. |
| AD–Seq–Vol / FT | P | **ready** | Full-geometry managed-GPU origin: 4,096 scenarios, 20 targets, 158.3 seconds, 2,636 MiB VRAM, no cap hit or nonconvergence; production pending. |

### 2.3 Supporting calibration and weekly evidence

At the first GROMM-eligible origin, 17 January 2023, the 200-strike affine market fit places 84.90% of observations inside their released spread. The rank-16 DLV innovation basis explains 95.15% of fitted innovation variance. The compatible-P projection explains only 0.633% of its target energy and its holdout loss improves by 0.473%, below the fixed 50% and 0.5% gates. The gain statistic is 9.54 and the daily risk-price norm is 0.102, but those two diagnostics do not override the failed acceptance gates. The artifact is therefore Q-only. Its float32 serialized densities miss unit mass by at most $1.964 \times 10^{-8}$; the loader now validates that residual against the stored precision, normalizes in float64 to $2.22 \times 10^{-16}$ maximum error, and rejects larger deviations. An authenticated four-path replay produced finite Q returns over 163 instruments and rejected P as required. These are calibration and implementation diagnostics, not hedge-performance results.

The weekly engine has 528 eligible origins through 22 August 2025 and 13 common targets: ten five-day digitals, realized variance, a 95% end-of-day down-and-out ATM put, and a 105% end-of-day up-and-out ATM call. The common DLV transition test has maximum mass and martingale errors below $3.6 \times 10^{-14}$ across an eight-date check.

Daily, frozen-weekly, and restarted-weekly adapters exist for all model families, but no complete weekly comparison is yet reported. The production matrix contains 94 cases. Its 28 ready SANOS–LV and historical-SANOS CPU cases are queued behind the active daily CPU runs; four generated-model cases await their production gate, and 62 GROMM or generated-model cases remain explicitly blocked rather than being launched without complete evidence. For Historical SANOS, real first-origin frozen-weekly gates cover all 13 targets: entropy at $\lambda = 10$ and power-exponential at $\lambda = 1$ both have zero cap hits and zero nonconverged fits. These are implementation gates, not weekly performance estimates.

## 3 Model Catalogue

### 3.1 SANOS–LV

SANOS supplies the common market interpolation and digital price [2]. DLV local volatility supplies the model price and Greeks. From each trailing 252-day window, the method estimates one front-spot-projected DLV factor and seven residual PCA factors. Central DLV-squared bumps give factor vegas; a linear program selects the minimum predictable-cost option portfolio that exactly neutralizes them, and spot removes residual delta. One-day claims are delta-only; three-day claims use at most eight option hedges. The method is deterministic, sparse, and the strongest completed baseline.

### 3.2 Historical SANOS

Historical SANOS reconstructs 252 same-coordinate option and target returns from released SANOS states [2] and uses those empirical rows directly as scenarios. It supports all four risk criteria in the deep-hedging sense of Buehler et al. [1]. Long, short, and zero policies determine indifference prices; a separate drift-neutral policy gives the executable pure hedge. Mean–variance uses exact cost-lasso coordinate descent, mean–volatility uses an exact conic program with an explicit recession test, and entropy and power-exponential objectives use an exact positive/negative action split for linear trading costs, with analytic gradients and an explicit KKT convergence gate. For weekly paths, entropy and power-exponential use exact conic epigraphs for scenario-dependent turnover costs plus the declared quadratic fitting ridge; reported certainty equivalents exclude that ridge. Only an optimal conic certificate passes the numerical gate. The model is inexpensive and fully rolling. Mean–variance at $\lambda = 0.1$ is complete on the current cohort with zero cap hits and zero nonconvergence; the remaining objective grid is still running.

### 3.3 Robust covariance

This baseline deliberately avoids a SANOS scenario generator. It replays the current relative option book over 252 historical public transitions, fits robust scales and a regularized Tyler scatter matrix [3, 5], and estimates each target cross-moment separately by robust regression. Expected returns are HAC-screened in covariance eigenportfolios. Mean–variance and mean–volatility hedges use spread-weighted lasso costs; the latter is checked analytically for recession directions. The implementation gate passes. Mean–variance at $\lambda \in \{0.1, 1, 10\}$ is complete on the current cohort; mean–volatility and the remaining objective grid are still running.

### 3.4 GROMM

GROMM is the native generative risk-neutral option-market model: a 200-strike JAX market fit, 16-factor AR(1) log-DLV-squared dynamics, independent 5% Black background, and eight Brownian substeps per business day. Dated artifacts are bound to exactly 252 public transitions and must pass compatible-P drift gates. The first rolling fit clears the market-fit threshold but rejects the compatible P measure, so it supplies Q scenarios only. Bounded float32 decode normalization and authenticated Q replay now pass. An affine-plus-deep LSTM supports all four objectives; whenever P is accepted, the executable pure policy uses conditional Q-shadow gains at each realized P state, removing predictable physical drift without fitted holdout centering. Full rolling production remains outstanding. A separate conventional Q comparator prices by risk-neutral expectation and computes spot delta plus eight contemporaneous DLV-factor vegas from the same stochastic GROMM state using fixed-seed common random numbers [ ].

### 3.5 VolGAN

VolGAN is a historical-P surface generator on a fixed $9 \times 11$ Black-IV grid reconstructed from public SANOS states. It conditions on two lagged returns, trailing 21-day realized volatility, and the current surface. The published two-layer Softplus architecture uses 32 noise dimensions, a 16-unit hidden layer, and strike/maturity gradient penalties. Weekly paths recursively update surface and return state. The implementation is project-native and uses no proprietary surfaces. Smoke, restart, and rolling warm-start gates pass; full production does not yet exist [6].

### 3.6 AD–Seq–Vol / FT

AD–Seq–Vol is a causal sequence diffusion model on 22-day histories of the same $9 \times 11$ surface plus forward-adjusted returns. Its eight-layer, width-256, four-head transformer predicts the clean sequence through a 200-step cosine DDPM. AD–Seq–Vol–FT freezes the EMA base and trains a zero-initialized rank-four adapter with calendar, call-spread, and butterfly penalties. Both are historical-P generators and supply indifference prices rather than Q expectations. Exact restart, rolling warm-start, daily/weekly adapters, and one full-geometry GPU origin pass; production and batch-invariance remain pending. The architecture and published comparison are due to Han et al. [ ].

**Provenance.** The current cohort's public dataset manifest begins `58741552e0f6`; its target-notional identity begins `1.1:fd6839a0`. The SANOS–LV results SHA-256 begins `9e4252f60540`. The covariance results SHA-256 values are `5ec474dd...`, `17cfc415...`, and `47ff392b...` for $\lambda = 0.1, 1, 10$, respectively. The Historical-SANOS mean–variance $\lambda = 0.1$ results SHA-256 begins `4197864e...`. Every headline number is derived from an immutable `hedging-daily-results-v2` artifact and its fully reconciled dated records.

## A Target-Level Out-of-Sample Objectives

The target-level appendix is intentionally withheld from this interim build. It will be generated only after the current-cohort historical-SANOS and remaining covariance policies complete, so the per-model, per-target, and per-objective tables cannot silently mix superseded target quantities with the current benchmark cohort. Objectives will be evaluated on unscaled inception-spot-normalized economic loss before the final scalar is converted to basis points.

## References

[1] Hans Buehler, Lukas Gonon, Josef Teichmann, and Ben Wood. Deep hedging. *Quantitative Finance*, 19(8): 1271–1291, 2019. doi: 10.1080/14697688.2019.1571683.

[2] Hans Buehler, Blanka Horvath, Anastasis Kratsios, Yannick Limmer, and Raeid Saqur. SANOS: Smooth arbitrage-free non-parametric option surfaces, 2026. URL https://arxiv.org/abs/2601.11209.

[3] Yilun Chen, Ami Wiesel, and Alfred O. Hero. Robust shrinkage estimation of high-dimensional covariance matrices. *IEEE Transactions on Signal Processing*, 59(9):4097–4107, 2011. doi: 10.1109/TSP.2011.2138698.

[4] Yinbin Han, Jack Yuxiang Zhang, Manuel Torres, Fernando Acero, and Renyuan Xu. Diffusion models for dynamic volatility surface generation and data-driven hedging, 2026. URL https://arxiv.org/abs/2609.13402.

[5] David E. Tyler. A distribution-free $M$-estimator of multivariate scatter. *The Annals of Statistics*, 15(1):234–251, 1987. doi: 10.1214/aos/1176350263.

[6] Milena Vuletić and Rama Cont. VolGAN: A generative model for arbitrage-free implied volatility surfaces. *Applied Mathematical Finance*, 31(4):203–238, 2024. doi: 10.1080/1350486X.2025.2471317.

[7] Magnus Wiese, Hans Buehler, and Blanka Horvath. GROMM: Generative risk-neutral option market models, 2026. Working paper.